Blockchain Consensus for Dynamic Robot Swarm Networks

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Abstract

Blockchain technology has significant potential to address key challenges in decentralized multi-robot systems, which require reliable distributed decision-making, fault tolerance, and secure communication. However, widely used consensus protocols such as Proof-of-Work (PoW) and Proof-of-Authority (PoA) do not account for the highly dynamic topologies of robot swarm networks, leading to inefficient block propagation and increased fork rates.
To address this limitation, this project presents a proof-of-concept for a novel consensus algorithm, Connectivity-aware-PoA (C-PoA). By prioritizing well-connected nodes as block producers, C-PoA aims to improve block propagation and overall data dissemination while maintaining low computational overhead.
The protocol is evaluated in simulation against PoW, PoA, and a randomized control protocol (R-PoA) under varying robot population sizes at constant density. Results indicate faster block propagation and more efficient block production. Although promising, these findings are not yet statistically validated and are based on scenarios that do not fully represent real-world swarm conditions.

1 Introduction

Robot swarms are increasingly used to address complex tasks such as exploration, mapping, and object localization through decentralized cooperation among simple agents. Their distributed nature enables robustness, scalability, and adaptability in dynamic environments. However, maintaining a consistent global state remains a fundamental challenge due to intermittent connectivity and constantly changing network topologies.

Blockchain technology has been proposed as a promising solution to these challenges, as it enables secure, transparent, and decentralized data management [Marco et al., 2024]. In particular, consensus mechanisms allow distributed agents to agree on a shared state without relying on a central authority. Widely

used mechanisms such as Proof-of-Work (PoW) and Proof-of-Authority (PoA) have been successfully applied in various domains, including recent work in swarm robotics [Van Calck et al., 2023, Strobel et al., 2023, Pachecol et al., 2025].

However, these consensus algorithms were originally designed for relatively stable network environments and do not explicitly account for the highly dynamic and frequently partitioned topologies of robot swarm networks[Marco et al., 2024]. As a result, their performance in such settings is limited, often leading to inefficient block propagation, increased fork rates, and reduced scalability.

To address these limitations, this work proposes Connectivity-aware-PoA (C-PoA). By prioritizing well-connected nodes, C-PoA aims to improve block dissemination and increases the likelihood that produced blocks are accepted by the network. In contrast to traditional approaches, this strategy directly adapts to dynamic connectivity conditions while maintaining low computational overhead.

To evaluate whether this connectivity-aware approach leads to measurable improvements, C-PoA is implemented within the Toychain framework [Alexandre et al., 2024] and assessed in simulation alongside PoW, PoA, and a randomized control protocol (R-PoA). By isolating the block producer selection mechanism as the primary variable, this study enables a controlled comparison of the different approaches under varying system sizes.

While the results are not statistically conclusive, they provide initial evidence that connectivity-aware block producer selection might improve consensus performance in robot swarms.

The remainder of this paper is structured as follows. Section 2 introduces the considered consensus protocols and their implementation. Section 3 describes the experimental setup and evaluation methodology. Section 4 presents and analyzes the experimental results. Finally, Section 5 provides concluding remarks, and Section 6 addresses the limitations of this study and proposes avenues for future research.

2 Methods

The consensus algorithms considered in this work are implemented within the Toychain Project, a lightweight blockchain framework designed for research in swarm robotics and are available at: https://github.com/donatienLeray/toychain. While PoW and PoA are already implemented and described in the Toychain technical report [Alexandre et al., 2024], they are revisited here to provide a detailed understanding of their operational characteristics and inherent limitations in dynamic network environments. This analysis forms the foundation for the design and justification of the proposed Connectivity-aware-PoA consensus protocol.

To ensure a fair comparison between consensus mechanisms, all protocols considered in this work share the same underlying blockchain structure, transaction

handling, and state update mechanisms. The only modified component is the block producer selection process. By isolating this mechanism as the primary variable, differences in performance can be attributed directly to the choice of consensus strategy.

2.1 Proof-of-Work

In Toychain, the Proof-of-Work consensus protocol follows the classical mining paradigm, in which nodes compete to produce valid blocks by solving a computationally intensive cryptographic puzzle, as originally introduced by Nakamoto [2008]. This mechanism ensures that block creation requires the expenditure of computational resources, thereby providing security against malicious behavior and enabling decentralized agreement on the blockchain state.

In the protocol a `MINING_DIFFICULTY` specifies the target difficulty that a block hash must satisfy in order to be considered valid. This parameter directly influences the expected time required to produce a block, as higher difficulty values require more iterations to find a suitable solution. Additionally, the implementation includes a `trust` flag, which allows nodes to bypass recomputation of the state stored in received blocks and instead accept it as valid. This option reduces computational overhead, which is particularly beneficial in simulation environments or resource-constrained systems.

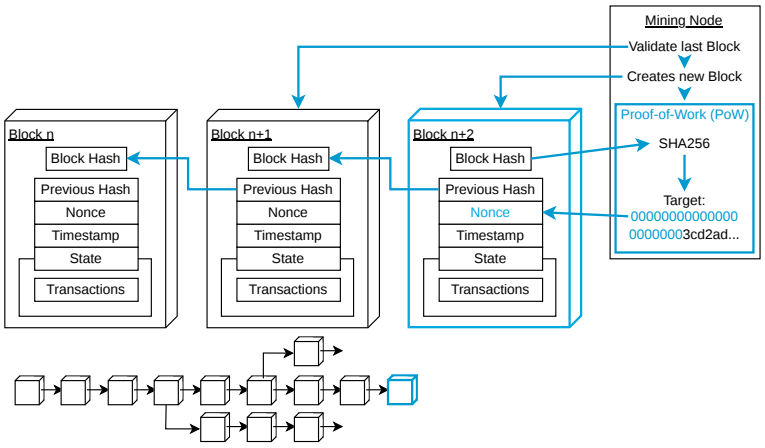


Figure 1: Proof-of-Work mining workflow: nodes iteratively vary a nonce until the block hash satisfies the difficulty target, then append and propagate the block.

A candidate block is constructed by aggregating transactions from the mempool, assigning a timestamp, and computing the state. The node then iteratively updates a nonce and recomputes the block hash until it satisfies the difficulty threshold. During this process, the block is continuously updated with the latest mempool contents and timestamp to reflect the network state. Once a valid hash is obtained, the block is appended to the local blockchain and propagated.

Each block stores not only its individual difficulty but also the sum of difficulties of all preceding blocks. When a node encounters multiple candidate chains, it selects the one with the highest total difficulty as the canonical chain. This approach ensures that chains representing greater cumulative computational effort are preferred.

Before accepting a received chain or chain segment, nodes perform a verification procedure to ensure compliance with the PoW rules. This process involves several checks.

1. The structural integrity of the chain is verified by ensuring that each block correctly references the hash of its predecessor.
2. Timestamps are validated to guarantee a strictly increasing temporal order.
3. Each block is checked against the required difficulty constraint to confirm that the Proof-of-Work condition has been satisfied.
4. The internal consistency of each block is verified by recomputing the state from its transactions and ensuring that it matches the state recorded in the block.

Only if all these conditions are met is the chain considered valid and accepted by the node.

Since real Proof-of-Work requires substantial computational effort, which is impractical in the context of simulations Toychain implements a simplified stochastic approximation of mining `VirtualMining`. Instead of repeatedly hashing until a valid solution is found, block production is modeled as a probabilistic process. Concretely, at each attempt, a random value is sampled and compared against a threshold determined by the node's computational capability and the global difficulty parameter:

1. A random difficulty value is drawn uniformly.
2. This value is compared to the product of the node's hash power (`ROBOT_HASHPOWER`) and the global `MINING_DIFFICULTY`.
3. Only if the sampled value exceeds this threshold does the node "win" the mining process and produce a block.

This mechanism effectively simulates the probabilistic nature of PoW without incurring the computational cost of repeated hashing. Nodes with higher hash power have a proportionally higher probability of producing blocks, preserving the fundamental behavior of PoW while remaining computationally lightweight.

When considering robot swarms, PoW presents fundamental limitations. Such systems require three key properties:

(i) sufficient security to prevent malicious manipulation.

(ii) low computational overhead due to hardware constraints.

(iii) fast block production to enable timely data propagation across the network.

In practice, PoW can only satisfy at most two of these requirements simultaneously. If the mining difficulty is set high, the system achieves strong security, as block production becomes hard to manipulate. However, this either requires significant computational resources or leads to increased block production times, both of which are undesirable in robot swarms. Conversely, lowering the difficulty improves block production speed and reduces computational demands, therefore reducing resilience against attackers, as blocks become easier to forge.

These limitations motivate the transition toward more efficient approaches such as Proof-of-Authority and, ultimately, the connectivity-aware-PoA protocol proposed in this work.

2.2 Proof-of-Authority

Proof-of-Authority [Szilágyi, 2017] has emerged as a state-of-the-art consensus protocol in blockchain-based robot swarms, primarily due to its ability to provide efficient and low-cost consensus without the computational burden associated with PoW. By replacing competitive mining with a controlled set of authorized block producers, PoA enables fast and predictable block generation, which is essential for time-sensitive coordination and data exchange in robotic swarms. As a result, PoA has been widely adopted in research on blockchain-enabled swarm robotics, such as Pacheco et al. [2025], Strobel et al. [2023], Pacheco et al. [2020].

In Toychain the set of authorized signers is defined in the genesis block, allowing the determination of eligible block producers at protocol initialization. Block production in PoA is governed by the `BLOCK_PERIOD`, which specifies the minimum time interval between consecutive blocks rather than enforcing strictly discrete rounds. A preferred signer is selected using a round-robin scheduling mechanism, ensuring a fair and deterministic rotation among all authorized nodes. Each signer waits for the duration of the block period before attempting to produce a block. If a node is not the designated in-turn signer, it waits an additional delay proportional to its position in the rotation, i.e., a multiple of `DELAY_NOTURN`, creating a cascading effect in which exactly one signer becomes eligible to produce a block at each step. Importantly, if a node receives a newly propagated block during its waiting period, the waiting time is reset according to the timestamp of that block. Only if the waiting time elapses without receiving a new block does the node proceed to create and broadcast its own block.

After this waiting period, a signer constructs a block by including all transactions currently present in its mempool and assigns a difficulty value that reflects its role in the round. Specifically, blocks produced by the in-turn signer are assigned a higher difficulty (`DIFF_INTURN`), while blocks produced by out-of-turn signers receive a lower difficulty (`DIFF_NOTURN`). These difficulty values do not represent computational effort, but instead act as a priority mechanism that influences chain selection.

Chain verification follows principles similar to those in PoW. Nodes validate the structural integrity of the chain by checking parent hashes, ensure that timestamps respect the minimum interval defined by `BLOCK_PERIOD`, verify that difficulty values are correctly assigned depending on the signer's role, and confirm that the state is consistent with the transactions contained in each block.

While PoA significantly reduces computational overhead, it introduces a fundamental trade-off in terms of decentralization. Since only a subset of nodes is authorized to produce blocks, the protocol relies on a certain level of trust in these participants. In research settings, this limitation is often mitigated by allowing all nodes in the network to act as authorized signers, thereby preserving decentralization.

To prevent any single node from dominating block production, an additional safeguard is introduced that guarantees that no node can create the majority of blocks over time. Each signer is only allowed to produce a block once within a window of $\lfloor N/2 \rfloor + 1$ blocks, where N is the total number of authorized signers. This mechanism prevents a single node from continuously producing blocks and ensures a more balanced distribution of block production rights, thereby improving resilience against faulty or malicious participants.

Despite these improvements, PoA remains fundamentally agnostic to the underlying network topology. In highly dynamic robot swarm networks, where communication links are intermittent, and network partitions are frequent, this can result in inefficient block propagation and increased fork rates [Marco et al., 2024].

2.3 Connectivity-aware-PoA

The discussed limitations of PoA motivate the need for a consensus mechanism that dynamically adapts to network conditions. This forms the basis for the proposed Connectivity-aware-PoA (C-PoA) protocol, which extends the PoA approach by replacing the static round-robin selection with a connectivity-aware selection of block producers.

The core idea is that, in distributed and dynamic networks, nodes with higher connectivity have a greater probability of rapidly disseminating information across the network. By prioritizing such nodes as block producers, C-PoA increases the likelihood that newly created blocks are quickly propagated and adopted by other participants, thereby reducing propagation delays and minimizing the occurrence of competing forks.

Contract

For C-PoA to operate, nodes must maintain a shared view of the network topology and derive a quantitative measure of their connectivity. This is achieved through a smart contract that records peer interactions and computes a connectivity index for each node. To this end, the inherited `state_variables` from `StateMixin` are extended with:

- `all_peers`: A dictionary tracking all observed peer relationships, represented as
  {`node_id:[(peer_id, timestamp),...],...`}
- `connectivity`: A dictionary storing the current connectivity value of each node,
  {`node_id:connectivity_value,...`}

To ensure the integrity of this information, updates to these variables are restricted to predefined smart contract functions:

- `AddPeer`: logs peer interactions by appending entries to `all_peers`
- `update_connectivity`: computes the connectivity value of each node by counting the number of unique mutual connections within a given time window (controlled by a decay parameter)

This design ensures that connectivity information is transparently recorded on the blockchain and cannot be arbitrarily manipulated by individual agents.

Block production

Block production in C-PoA as summarized in the Algorithm 1 follows the same timing mechanism as in PoA, including the cascading delay, reset behavior and $\lfloor N/2 \rfloor + 1$ restriction. However, the ordering of nodes is no longer static but dynamically determined by their connectivity.

At each block, the connectivity values stored in the state define a global ranking of all authorized signers. The node with the highest connectivity is selected as the preferred signer and is therefore allowed to produce a block immediately after the `BLOCK_PERIOD`. Nodes with lower connectivity must wait an additional delay proportional to their position in the ranking, i.e., $k \cdot$ `DELAY_NOTURN` for the node ranked at position k.

If multiple nodes have identical connectivity values, their relative order is determined through a deterministic randomization seeded by the previous block, ensuring consistency across all nodes.

Importantly, the connectivity ranking is already determined in the previous block, as each block stores the current connectivity state. This allows all nodes to independently compute the expected order of block producers.

In addition to determining the production order, the connectivity of the block producer is used as the block difficulty. As a result, blocks produced by highly connected nodes contribute more to the total difficulty of the chain. This design favors blocks that are more likely to be propagated quickly through the network, thereby improving consensus efficiency and reducing the likelihood of forks.

```
Data: self node
Result: update blockchain
while consensu is running do
    // Get last bock
    B_{n-1} ← node.chain[-1];
    connectivity ← B_{n-1}.state.connectivity;
    // Randomize to break ties
    connectivity ← randomize(connectivity, seed = B_{n-1}.hash);
    connectivity_ranking ← stable_sort_by_value(connectivity);
    // No node may forge more than once within ⌊N/2⌋ + 1 blocks
    N ← length(connectivity);
    n ← B_{n-1}.number + 1;
    if n − node.last_signed < ⌊N/2⌋ + 1 then
        return;
    end
    k ← get_position(node, connectivity_ranking);
    t ← current timestamp;
    t_old ← B_{n-1}.timestamp;
    // Preferred signer may forge immediately after BLOCK_PERIOD
    // Out-of-turn signing follows staggered delay, as in PoA
    if t − t_old − BLOCK_PERIOD < k · DELAY_NOTURN then
        return;
    end
    // Create new block
    B_n ← copy(B_{n-1});
    // apply transactions
    foreach transaction ∈ node.mempool do
        if transaction ∉ B_{n-1}.transactions then
            B_n.state.apply(transaction);
            B_n.transactions.add(transaction);
        end
    end
    // difficulty according to connectivity value
    B_n.difficulty ← connectivity[node] + 1;
    B_n.total_difficulty ← B_{n-1}.total_difficulty + B_n.difficulty;
    B_n.state.payout_block_reward(B_{n-1});
    // recalculate connectivity
    B_n.state.update_connectivity();
    // update all other block variable accordingly
    B_n.previous_hash ← B_{n-1}.hash;
    B_n.hash ← hassch(B_n.state, B_n.previous_hash;
    B_n.number ← n;
    B_n.timestamp ← t;
    B_n.signer ← node;
    node.lastsigned ← n
    // Append block and clear included transactions
    node.chain.append(B_n);
    node.mempool.clear();
    return;
end
```

Algorithm 1: *Connectivity-aware PoA block production with connectivity-based signer ranking, staggered delays, and difficulty proportional to node connectivity.*

Chain Verification

Chain verification in C-PoA extends the validation procedure of PoA by incorporating connectivity-based checks. Specifically, for each block, a node recomputes the connectivity values from the stored state and derives the corresponding ranking. It then verifies that:

- the block producer corresponds to the node assigned to that position in the ranking
- the waiting time implied by the ranking position has been respected
- the difficulty of the block matches the connectivity value of the producer

These checks ensure that block production strictly follows the connectivity-based ordering and that nodes cannot arbitrarily claim a more favorable position in the ranking.

2.4 Randomized-Proof-of-Autherity

To serve as a baseline for evaluating whether the connectivity-aware selection mechanism of C-PoA provides a measurable improvement over an unstructured approach, a control protocol is introduced in which the block producer is selected uniformly at random.

The Randomized-Proof-of-Autherity (R-PoA) protocol is implemented as a variant of C-PoA by adapting the `update_connectivity` function in the smart contract to perform no updates. As a result, all nodes maintain identical connectivity values throughout execution. Consequently, the connectivity-based ranking used for block producer selection is determined solely by the tie-breaking mechanism, which applies a deterministic random ordering seeded by the previous block.Since all nodes have equal connectivity, each block is assigned the same difficulty value, reducing fork resolution to a longest-chain rule based on block count.

It is important to note that the probabilistic behavior of block production differs fundamentally between PoW and R-PoA. In the simulated PoW implementation, each node independently attempts to produce a block with a probability determined by its local hash power, resulting in a set of parallel and independent Bernoulli trials. Consequently, multiple nodes may succeed within a similar time frame, increasing the likelihood of concurrent block creation and thus forks.

In contrast, R-PoA models block producer selection as a network-wide stochastic process, in which the candidate producer is determined from a uniformly random ordering derived from a shared seed in the previous block. Thus, nodes that agree on the current chain tip also agree on the next candidate producer. However, under delayed propagation or temporary forks, nodes may derive different candidates from local views of the chain. Compared to PoW, this still substantially reduces probabilistic contention, as block production is tied to the current blockchain state rather than independent local trials at every node.

3 Experimental evaluation

This section presents the experimental evaluation of the considered consensus protocols. It first describes the experimental setup, including the simulation environment, execution procedure, and data collection process. Subsequently, the metrics used for evaluation are introduced and their computation from the recorded data is outlined. Finally, the results are presented and analyzed to assess the performance of the different approaches.

3.1 Experiment Setup

The experiments are conducted using Toychain-ARGOS [Alexandre et al., 2024], an existing integration of the Toychain framework into the ARGoS multi-robot simulator [Pinciroli et al., 2012]. This setup enables the evaluation of blockchain-based consensus protocols in a simulated robot swarm. All code required to run the experiments and evaluate the results is available at:
https://github.com/donatienLeray/toychain-argos

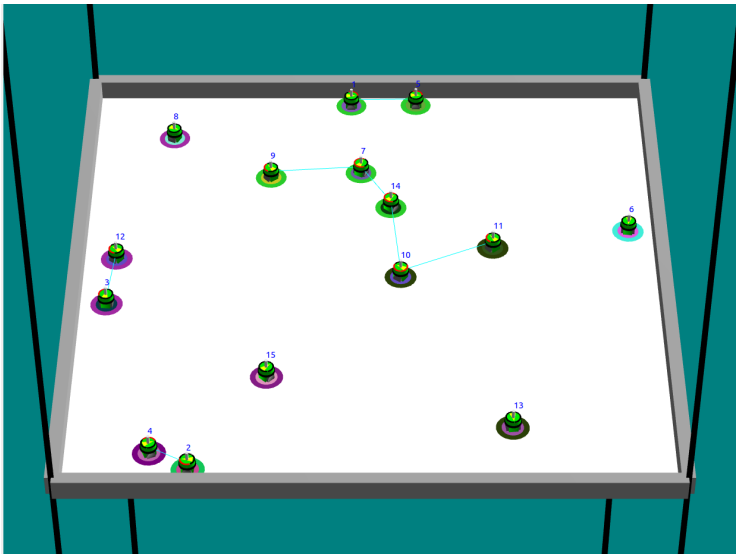


Figure 2: Simulation environment showing robots performing a random walk in the arena and communicating with peers. The color of the inner circle indicating the state of the memory pool while the color of the outer circle indicating the state of their chain.

The simulations are based on the provided "Hello World" scenario, in which robots operate in an empty rectangular arena and perform a random walk. During execution, robots periodically generate and broadcast transactions to neighboring peers, simulating simple communication and interaction behavior in a dynamic environment. The simulation uses a 2D physics engine, ensuring realistic movement behavior including collision avoidance between robots and with the arena boundaries.

At the beginning of each experiment, robots are uniformly distributed within a central region of the arena, with their orientations sampled from a Gaussian distribution. This results in randomized initial conditions while maintaining a controlled spatial distribution.

Experimental Parameters

The experiments are conducted under controlled conditions with a set of fixed and varying parameters.
Fixed parameters:

- Robot model: e-puck2
- Robot behavior: random walk with periodic transaction generation
- Communication range: 0.5 m
- Simulation ticks per second: 10 (control loop frequency)
- Block period: 1 s
- Agent density: $2\,\frac{\text{agents}}{\text{m}^2}$, maintained by scaling the arena size
- Simulation duration: 400 seconds (4000 ticks)
- Wheel noise: none
- Number of repetitions per configuration: 10

Varied parameters:

- Consensus protocol (PoW, PoA, C-PoA, R-PoA)
- Number of robots (5–25 in steps of 5)

For each consensus protocol, experiments are conducted for robot populations ranging from 5 to 25 in steps of 5. Each configuration is repeated 10 times. To ensure comparability, runs with the same number of robots and repetition index use identical initial positions.

Varying the number of robots allows the evaluation of scalability across different system sizes. After each run, the complete blockchain and the full log data of each robot are recorded and stored as CSV files for evaluation.

All recorded data is stored as CSV files and subsequently processed to compute the defined evaluation metrics.

3.2 Metrics

The following metrics are used to evaluate the performance of the considered consensus protocols.

Block Production Efficiency (BPE)
The percentage of produced blocks that are included in the main chain, indicating how many blocks are successfully adopted rather than abandoned.

Total Produced Blocks (TPB)
The total number of blocks produced by all agents during a simulation run, regardless of whether they are included in the main chain or discarded. This metric reflects the overall block production activity and indicates the level of contention and redundant work within the network.

Block Reception Delays (BRD)
The time required for a block to propagate through the network from its creation until it is received by other nodes, reflecting how quickly information can spread across the system.

Block Production Delays (BPD)
The time difference between the creation of consecutive blocks on the main chain, reflecting the rate at which confirmed state updates occur within the system.

Block Production Inequality (BPI)
To quantify how evenly block production is distributed among agents, we analyze the block production inequality (BPI). This metric is based on the Gini coefficient [Gini, 1912], a standard measure of inequality, where a value of 0 indicates perfectly equal participation among all nodes and a value of 1 indicates that a single node produces all blocks.
BPI is computed based on the number of blocks each agent contributes to the main chain, thereby capturing how concentrated block production is within the system. This provides a compact and comparable measure of fairness across different consensus mechanisms and system sizes.

4 Results

This section presents and analyzes the experimental results of the considered consensus protocols. The results are interpreted with respect to the defined metrics to evaluate scalability and the effectiveness of the different approaches.

Block Production Efficiency (BPE)

To assess how well the different consensus algorithms scale with increasing system size, we analyze the block production efficiency (BPE). As system size increases and communication becomes more challenging, BPE provides insight into how effectively each protocol maintains consistent block inclusion under increasing network constraints.

As shown in Fig. 3, C-PoA exhibits a noticeably higher variance at a population size of 5 agents. This behavior is not inherent to the connectivity-based selection itself but is caused by the $\lfloor N/2 \rfloor + 1$ constraint. When all agents have produced a block within the last $\lfloor N/2 \rfloor + 1 = 3$ blocks while simultaneously maintaining chains of equal total difficulty. No agent is allowed to produce a new block, and no alternative chain is preferred, resulting in a deadlock. This phenomenon was observed in approximately 10% of the C-PoA runs with 5 agents. While this

situation is theoretically possible for PoA and R-PoA and every population size, it did not occur with in the conducted experiments.

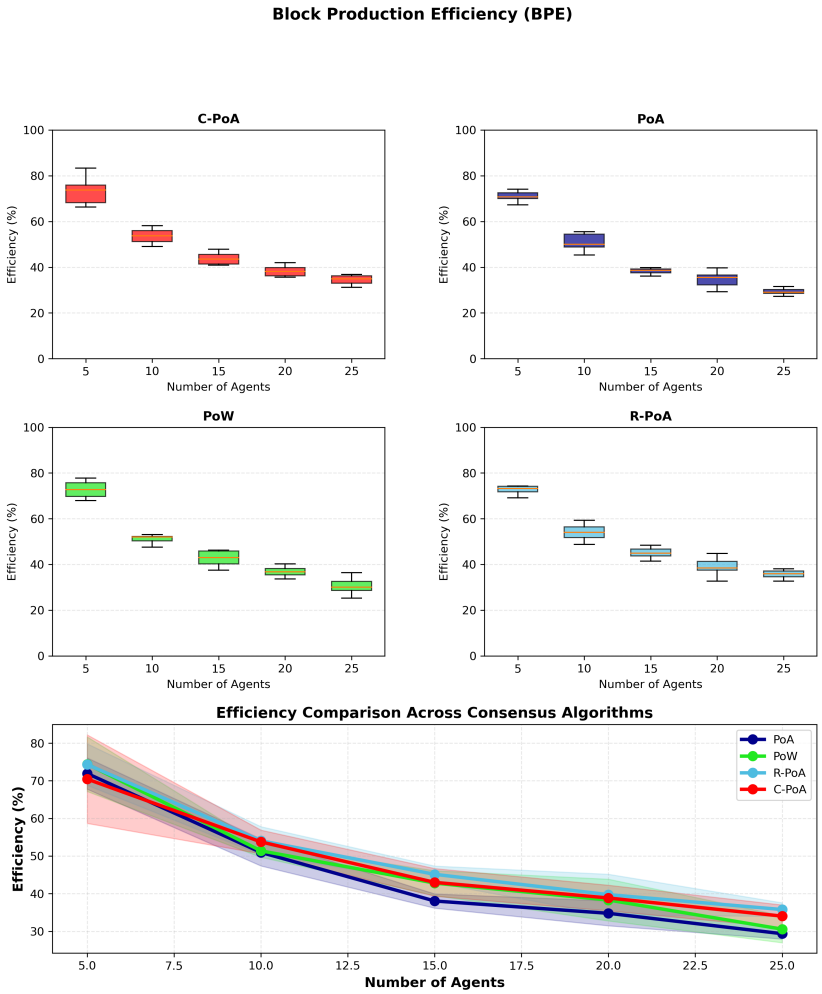


Figure 3: Block production efficiency across swarm sizes (10 runs per configuration): R-PoA achieves slightly higher efficiency, while C-PoR shows competitive scaling with less variance.

Considering the scaling behavior, C-PoA shows a slightly higher efficiency than PoA and PoW across increasing population sizes. This aligns with the expectation that prioritizing well-connected nodes improves block propagation and reduces the likelihood of blocks being discarded. However, the observed differences are relatively small, and their statistical significance remains to be established.

Interestingly, R-PoA appears to scale slightly better than the other protocols. A possible explanation is that the uniform random selection occasionally allows nodes outside the deterministic ordering to produce blocks, which can mitigate

unfavorable synchronization effects and reduce the likelihood of temporary contention or deadlocks. However, given the higher variance observed in R-PoA, this effect should be interpreted with caution and cannot be considered statistically conclusive.

Total Produced Blocks (TPB)

BPE alone does not capture the overall level of block production within the system. To obtain a more complete understanding of protocol behavior, it is necessary to consider the total number of produced blocks (TPB) alongside BPE. Only the combination of these two metrics reflects the overall activity in the network and provides insight into the level of contention and redundant work generated by the different consensus mechanisms.

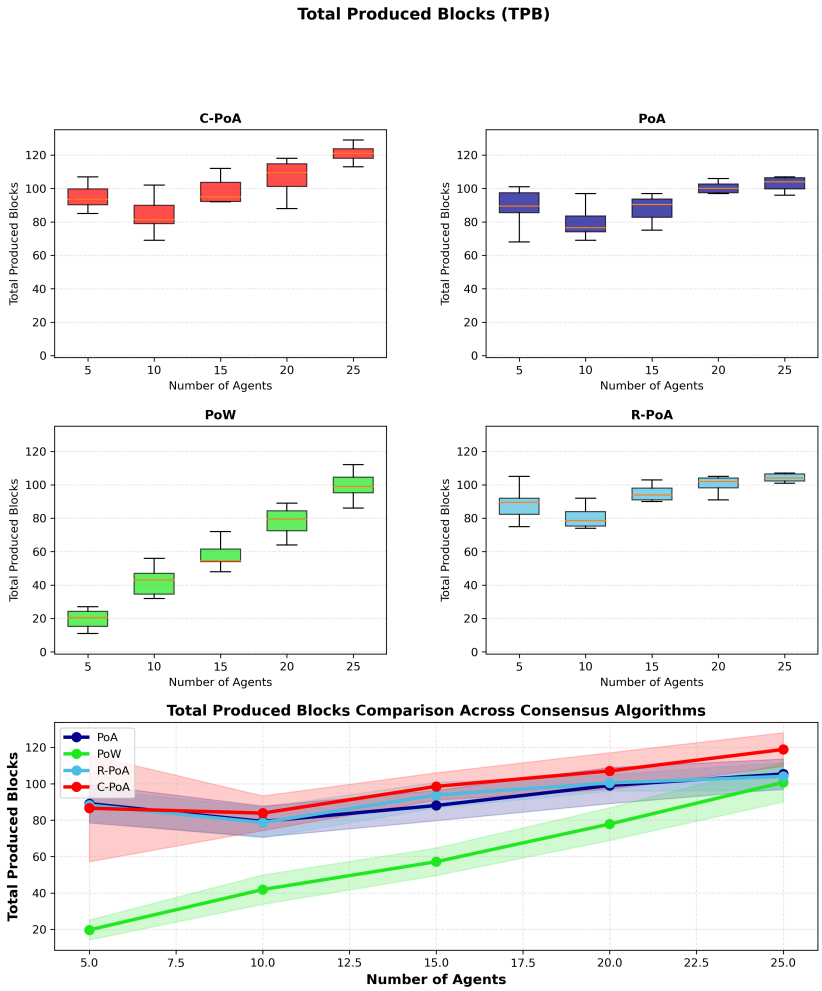


Figure 4: Total produced blocks across swarm sizes (10 runs per configuration): C-PoA consistently generates the highest number of blocks, indicating increased network activity.

As shown in Fig. 4, C-PoA consistently produces a higher number of blocks compared to the other consensus mechanisms, particularly as the number of agents increases. This indicates a higher level of overall activity, which can be attributed to the connectivity-based selection favoring nodes that are well-positioned to propagate blocks quickly.PoA and R-PoA exhibit similar trends with increasing system size, but with consistently lower total block production.

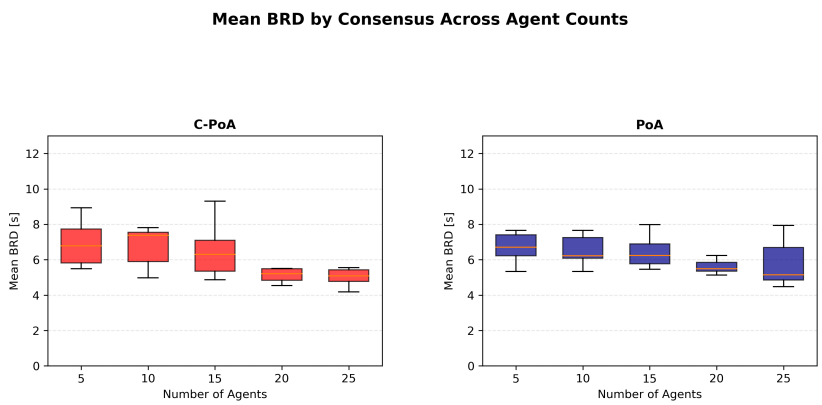
It should be noted that PoW is not directly comparable to PoA, C-PoA, and R-PoA regarding TPB due to its fundamentally different block production mechanism. In PoW, the total number of produced blocks is largely determined by the chosen difficulty parameter, and lowering the target difficulty would increase TPB. However, as discussed previously, this would come at the cost of reduced security and would likely decrease efficiency due to a higher rate of competing blocks, an effect that would require further experimental validation.

Block Reception Delays (BRD)

To evaluate how efficiently blocks propagate through the network, we analyze the block reception delays (BRD), which capture the time required for blocks to be received by other nodes after their creation.

As shown in Fig. 5, the mean block reception delay generally decreases with increasing system size across all considered consensus mechanisms. Since agent density is kept constant, this trend can likely be attributed to the increased number of communication links, which provide more alternative paths for block dissemination.However, the overall high variance range, makes it difficult to draw firm conclusions about scalability or underlying trade-offs.

Furthermore, BRD alone is not a sufficient indicator of overall consensus performance, as faster propagation does not necessarily imply that blocks are successfully incorporated into the main chain. Therefore, this metric must be interpreted in conjunction with others that capture effective state progression.



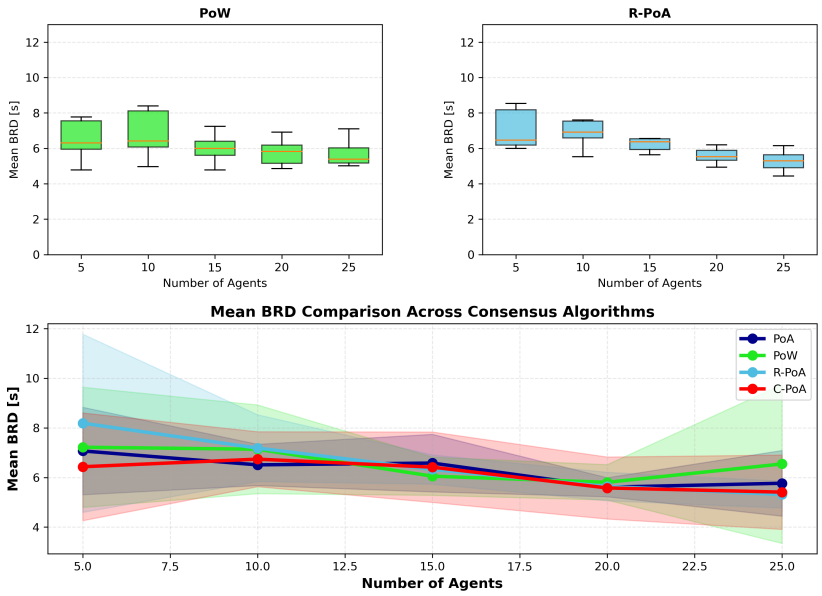


Figure 5: Block reception delay across swarm sizes (10 runs per configuration): delays decrease with scale across protocols, with high variance limiting clear differentiation.

Block Production Delays (BPD)

To quantify how frequently the global system state is updated, we analyze the block production delays (BPD).

To evaluate how efficiently blocks propagate through the network, we analyze the block reception delays (BRD), which capture the time required for blocks to be received by other nodes after their creation.

As shown in Fig. 6, C-PoA consistently achieves the lowest block production delays across all evaluated system sizes, indicating faster updates of the global state compared to the other consensus mechanisms. This behavior can be attributed to the connectivity-aware selection strategy, which promotes the production of blocks that are more likely to be rapidly propagated and accepted by the network.

PoA and R-PoA exhibit similar trends, with delays increasing as the system size grows, suggesting that coordination overhead and contention begin to limit the rate at which blocks are successfully added to the main chain. PoW, in contrast, shows significantly higher delays, particularly at smaller system sizes, reflecting the probabilistic nature of block creation and the reduced likelihood of frequent successful block generation.

Furthermore, C-PoA appears to approach an asymptotic behavior around 10 seconds as the number of agents increases. This suggests that the protocol converges to a stable operating regime in which block production and propagation

remain well-balanced, even as the system scales.

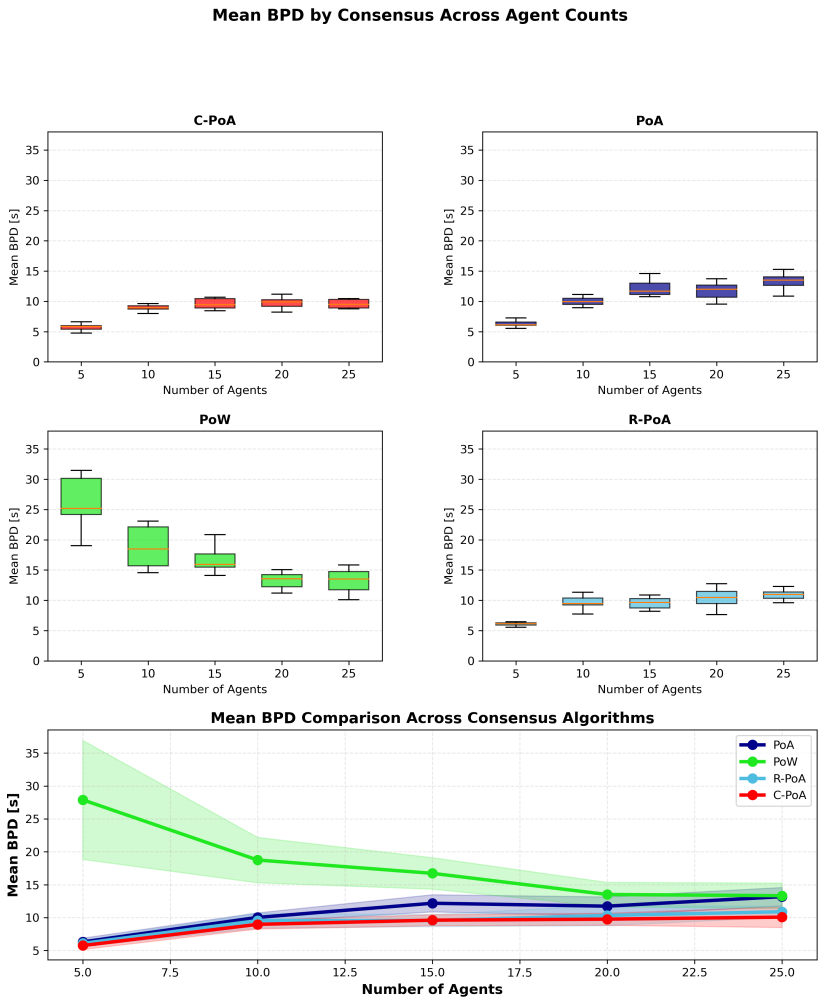


Figure 6: Block production delay across swarm sizes (10 runs per configuration). C-PoA yields the lowest delays, enabling faster global state updates.

Overall, C-PoA demonstrates superior performance in terms of effective state update frequency. However, as with the other metrics, the observed differences should be interpreted with caution, as their statistical significance has not yet been established.

Block Production Inequality (BPI)

To ensure that all nodes have a fair opportunity to contribute to the global system state, it is necessary to evaluate how evenly block production is distributed among participants. A consensus mechanism that allows only a small subset of

nodes to dominate block production reduces effective decentralization, as other nodes become increasingly reliant on these dominant participants. Therefore, block production inequality (BPI) can also be interpreted as a measure of how decentralized a consensus algorithm is in practice.

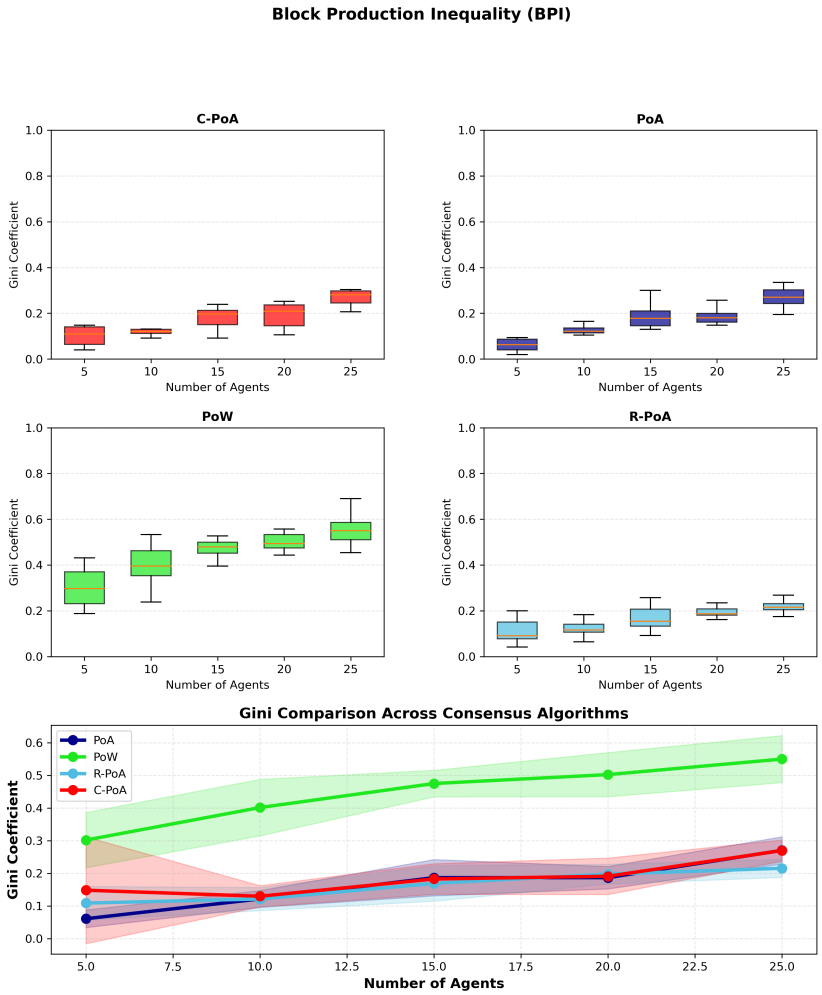


Figure 7: Block production inequality across swarm sizes (10 runs per configuration): PoW exhibits high inequality, while R-PoA maintains the most uniform block distribution.

As shown in Fig. 7, PoW exhibits a significantly higher level of inequality compared to the other consensus mechanisms. In this case, block production is highly concentrated among a few nodes, while many agents contribute little or not at all. This behavior can be largely attributed to the absence of an explicit constraint limiting how frequently a node may produce blocks, such as the $\lfloor N/2 \rfloor + 1$ rule used in the other protocols. Without such a restriction, nodes that successfully extend the main chain gain a persistent advantage, leading to a reinforcing effect in which a small subset of participants dominates block

production.

It is important to note that such a constraint cannot be directly applied to PoW. Since PoW operates in a permissionless setting, participants can create multiple identities (Sybil attacks), effectively bypassing any rule that limits block production per node.

As expected, R-PoA achieves the lowest inequality among all evaluated protocols. In this case, the combination of uniform random selection and the $\lfloor N/2 \rfloor + 1$ constraint prevents any node from gaining a systematic advantage, resulting in an almost uniform distribution of block production across the network.

5 Conclusion

This work introduced Connectivity-aware Proof-of-Authority (C-PoA), a novel consensus mechanism designed to address limitations of existing approaches in dynamic multi-robot systems. By incorporating network connectivity into the selection of block producers, C-PoA aims to improve block propagation and overall consensus efficiency in environments characterized by intermittent communication and evolving network topologies.

Although the experimental results are not statistically significant, they provide consistent indications that C-PoA offers advantages over the evaluated protocols. In particular, C-PoA maintains comparable or slightly improved block production efficiency while consistently generating a higher number of blocks, enabling more frequent updates of the global system state aka consents. Collectively, these findings indicate that connectivity-aware block producer selection can contribute to faster consensus formation across the network.

Importantly, C-PoA does not exhibit major drawbacks compared to existing approaches, except for a rare deadlock scenario observed under specific conditions due to the $\lfloor N/2 \rfloor + 1$ constraint. While infrequent, this limitation highlights the need for further investigation and potential mitigation strategies.

Additionally, the randomized control protocol (R-PoA) consistently outperforms standard PoA at higher agent counts across several metrics. This suggests that introducing controlled randomness into block producer selection may help mitigate synchronization effects and improve overall system performance.

Overall, the results indicate that connectivity-aware and randomized block producer selection strategies can provide meaningful benefits over traditional consensus mechanisms such as Proof-of-Authority and Proof-of-Work in dynamic network environments. While further validation is required—particularly with respect to statistical significance and more realistic deployment scenarios—both C-PoA and R-PoA represent promising directions for future research.

6 Future Work

While the presented results suggest that Connectivity-aware Proof-of-Authority (C-PoA) is a promising approach, several aspects require further investigation before it can be considered a robust and practically viable consensus mechanism.

First, the deadlock scenario arising from the $\lfloor N/2 \rfloor + 1$ constraint must be addressed. Although this issue was observed only rarely, its resolution is essential to ensure reliable operation and suitability for real-world deployment.

Second, the current evaluation is limited to a simplified random walk scenario. To better assess its applicability, C-PoA should be evaluated in more complex multi-robot system tasks, such as coordinated foraging, exploration, and mapping, where communication patterns and network dynamics differ substantially. Such experiments would provide a more realistic assessment of its performance in practical multi-robot environments.

Third, C-PoA currently represents an extension of Proof-of-Authority rather than a fully independent consensus mechanism. In particular, its resilience to adversarial behavior remains unexamined. The protocol's reliance on the $\lfloor N/2 \rfloor + 1$ constraint to ensure fairness and prevent node dominance further limits its conceptual independence. Future work should therefore investigate whether this constraint can be relaxed or replaced without compromising correctness, as well as evaluate the protocol's robustness against adversarial scenarios, including malicious nodes and network manipulation. Establishing such properties is necessary to determine whether C-PoA can provide security guarantees comparable to established consensus mechanisms.

Finally, the performance of the randomized baseline (R-PoA) warrants further investigation. If the observed improvements persist under more rigorous and statistically significant evaluation, R-PoA may offer a practical enhancement over standard Proof-of-Authority without introducing additional complexity.

Overall, further research is required to validate the effectiveness, robustness, and generalizability of C-PoA in realistic settings and to explore the broader design space of connectivity- and randomness-based consensus mechanisms.

References

Pacheco Alexandre, Denis Ulysse, Zakir Raina, Strobel Volker, Reina Andreagiovanni, and Dorigo Marco. Toychain: A simple blockchain for research in swarm robotics, 2024. URL https://arxiv.org/abs/2407.06630.

Corrado Gini. *Variabilità e mutabilità: contributo allo studio delle distribuzioni e delle relazioni statistiche.[Fasc. I.]*. Tipogr. di P. Cuppini, 1912.

Dorigo Marco, Pacheco Alexandre, Reina Andreagiovanni, and Strobel Volker. Blockchain technology for mobile multi-robot systems. *Nature Reviews Electrical Engineering*, 1 (4):264–274, 2024. doi: 10.1038/s44287-024-00034-9. URL https://doi.org/10.1038/s44287-024-00034-9.

Satoshi Nakamoto. Bitcoin: A peer-to-peer electronic cash system, 2008. URL https://bitcoin.org/bitcoin.pdf. Accessed: 2024-12-04.

Alexandre Pacheco, Volker Strobel, and Marco Dorigo. A blockchain-controlled physical robot swarm communicating via an ad-hoc network. In *International Conference on Swarm Intelligence*, pages 3–15. Springer, 2020.

Alexandre Pacheco, Hanqing Zhao, Volker Strobel, Tarik Roukny, Gregory Dudek, Andreagiovanni Reina, and Marco Dorigo. Swarm oracle: Trustless blockchain agreements through robot swarms. *arXiv preprint arXiv:2509.15956*, 2025. URL https://doi.org/10.48550/arXiv.2509.15956.

Alexandre Pachecol, Sébastien De Vos[1], Andreagiovanni Reina, Marco Dorigo[1], and Volker Strobell. Securing federated learning in robot swarms using blockchain technology. In *Distributed Autonomous Robotic Systems: 17th International Symposium*, page 473. Springer Nature, 2025. URL https://doi.org/10.48550/arXiv.2409.01900.

Carlo Pinciroli, Vito Trianni, Rehan O'Grady, Giovanni Pini, Arne Brutschy, Manuele Brambilla, Nithin Mathews, Eliseo Ferrante, Gianni Di Caro, Frederick Ducatelle, Mauro Birattari, Luca Maria Gambardella, and Marco Dorigo. ARGoS: a modular, parallel, multi-engine simulator for multi-robot systems. *Swarm Intelligence*, 6(4):271–295, 2012.

Volker Strobel, Alexandre Pacheco, and Marco Dorigo. Robot swarms neutralize harmful byzantine robots using a blockchain-based token economy. *Science Robotics*, 8(79): eabm4636, 2023. doi: 10.1126/scirobotics.abm4636. URL https://www.science.org/doi/abs/10.1126/scirobotics.abm4636.

Péter Szilágyi. EIP-225: Clique Proof-of-Authority Consensus Protocol. Ethereum Improvement Proposals, no. 225, March 2017. https://eips.ethereum.org/EIPS/eip-225.

Ludéric Van Calck, Alexandre Pacheco, Volker Strobel, Marco Dorigo, and Andreagiovanni Reina. A blockchain-based information market to incentivise cooperation in swarms of self-interested robots. *Scientific Reports*, 13(1):20417, 2023. URL https://www.nature.com/articles/s41598-023-46238-1.